

DAEN/ISEN 427 Homework 2

Due date: Tue Sep 16, 2025 (before class)

Rubric: Full credit may be given for any reasonable Bayesian analysis.

1. From the following expressions, which one corresponds to the sentence: *The probability of being sunny given that it is June 14th of 1976?*

- (a) $p(\text{sunny})$
- (b) $p(\text{sunny} \mid \text{June})$
- (c) $p(\text{sunny} \mid \text{June 14th of 1976})$
- (d) $p(\text{June 14th of 1976} \mid \text{sunny})$
- (e) $p(\text{sunny, June 14th of 1976})/p(\text{June 14th of 1976})$

Answer:

2. Show that the probability of choosing a human at random and picking the Pope is not the same as the probability of the Pope being human. In the animated series Futurama, the (Space) Pope is a reptile. How does this change your previous calculations?

Answer:

3. Suppose the prior distribution and the posterior distribution are as in Figure 1:

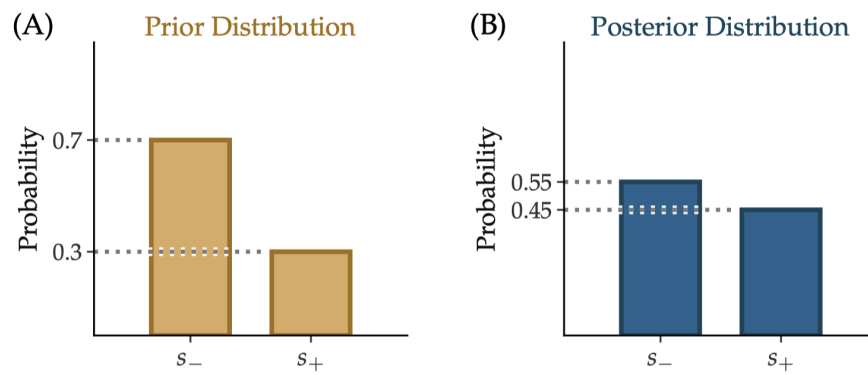


Figure 1: Prior and Posterior for Discrimination Task

(a) Calculate the likelihood ratio.

Answer:

(b) Does the sensory evidence alone (without the prior) indicate that the stimulus was s_+ or s_- ?

Answer:

(c) Do the prior and the likelihood favor the same alternative?

Answer:

4. Assume the likelihood function and a posterior distribution are Gaussian (see figure 2). Assume that the prior is also Gaussian. Which of the following statements is true? Explain.

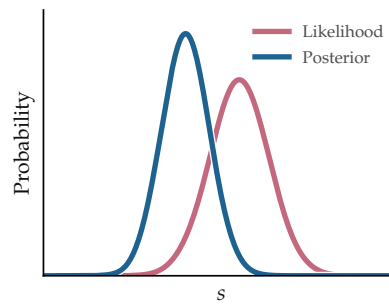


Figure 2: Gaussian distributions

- (a) The prior is centered to the left of the likelihood function and is narrower.
- (b) The prior is centered to the left of the likelihood function and is wider than it.
- (c) The prior is centered to the right of the likelihood function and is narrower than it.
- (d) The prior is centered to the right of the likelihood function and is wider than it.

Answer: